

RGCS Engineering Build Sheet — Coil and Pulse

Design CPD-925-001 · assembly PHY-BASIC-001 · mode am_key

Wire and coil estimates (model outputs)

wire	AWG 26 copper
wire length estimate (m)	62.8319
resistance estimate (ohm)	8.19826
DC current at voltage limit (A)	5.48897

Sideband table (carrier ± n·key)

order	lower (Hz)	upper (Hz)	note
1	3171	5021	
2	2246	5946	
3	1321	6871	

Pulse timing table

segment	description	f (Hz)	period (µs)	duty
carrier	base carrier 4096 Hz	4096	244.141	0.5
modulation	amplitude envelope at 925 Hz	925	1081.08	0.5

Modulation key

modulation key status: mathematical relation

Measurement plan

Measure actual coil resistance and driven waveforms before relying on any estimate above; record measured values beside the model outputs.

Claim boundary

This build sheet is an engineering plan and reproducibility record. Predictions are model outputs. Measurements decide.

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